

CS 70 — Discussion 2D

Problem 1: Baby Fermat

Assume a has a multiplicative inverse mod m . We prove the inverse can be written as $a^k \pmod{m}$ for some $k \geq 0$.

Part (a): The sequence $a, a^2, a^3, \dots \pmod{m}$ has repetitions

Each term of the sequence is an element of $\{0, 1, \dots, m-1\}$, so there are at most m distinct possible values. Consider the first $m+1$ terms $a^1, a^2, \dots, a^{m+1} \pmod{m}$. By the Pigeonhole Principle, $m+1$ terms mapping into m possible residues forces at least two terms to be equal: $a^i \equiv a^j \pmod{m}$ for some $i > j$. Hence the sequence has repetitions. $\square$

Part (b): If $a^i \equiv a^j \pmod{m}$ with $i > j$

Starting from $a^i \equiv a^j \pmod{m}$, write $a^i = a^{i-j} \cdot a^j$:

$$a^{i-j} \cdot a^j \equiv a^j \pmod{m}.$$

Since a has an inverse mod m , so does a^j . Multiplying both sides by $(a^j)^{-1}$:

$$a^{i-j} \equiv 1 \pmod{m}.$$

So $a^{i-j} \equiv 1 \pmod{m}$, where $i-j > 0$.

Part (c): Expressing the inverse as a power of a

From part (b), $a^{i-j} \equiv 1 \pmod{m}$. Pulling out one factor of a :

$$a \cdot a^{i-j-1} \equiv 1 \pmod{m}.$$

Thus a^{i-j-1} is the multiplicative inverse of a , so $k = i-j-1$. Since $i > j$, we have $i-j \geq 1$, so $k = i-j-1 \geq 0$ as required.

Problem 2: RSA Practice

Part (a): Finding the decryption key

With $p = 5$, $q = 11$, so $N = pq = 55$ and $(p - 1)(q - 1) = 4 \cdot 10 = 40$. The encryption key is $e = 9$. The decryption key is $d \equiv e^{-1} \equiv 9^{-1} \pmod{40}$.

Extended Euclidean on $\gcd(40, 9)$:

$$\begin{array}{ll} 40 = 4 \cdot 9 + 4 & 4 = 40 + (-4) \cdot 9 \\ 9 = 2 \cdot 4 + 1 & 1 = 9 + (-2) \cdot 4 \\ 4 = 4 \cdot 1 + 0 & \end{array}$$

Back-substituting:

$$1 = 9 - 2 \cdot 4 = 9 - 2(40 - 4 \cdot 9) = 9 \cdot 9 - 2 \cdot 40.$$

So $9 \cdot 9 \equiv 1 \pmod{40}$, giving $\boxed{d = 9}$.

Part (b): Decrypting the received ciphertext

The receiver gets $y = 4$. The original message is $x = y^d = 4^9 \pmod{55}$. By repeated squaring:

$$4^2 \equiv 16, \quad 4^4 \equiv 16^2 = 256 \equiv 36, \quad 4^8 \equiv 36^2 = 1296 \equiv 31 \pmod{55}.$$

Then $4^9 = 4^8 \cdot 4 \equiv 31 \cdot 4 = 124 \equiv 14 \pmod{55}$. So the original message was $\boxed{14}$.

Part (c): Verifying by re-encoding

Re-encrypt $x = 14$ using $E(x) = x^e = 14^9 \pmod{55}$:

$$14^2 \equiv 196 \equiv 31, \quad 14^4 \equiv 31^2 = 961 \equiv 26, \quad 14^8 \equiv 26^2 = 676 \equiv 16 \pmod{55}.$$

Then $14^9 = 14^8 \cdot 14 \equiv 16 \cdot 14 = 224 \equiv 4 \pmod{55}$. This returns the original ciphertext 4, confirming the answer. ✓

Problem 3: RSA with Three Primes

Modified scheme. Let $N = pqr$ for distinct primes p, q, r . The scheme is:

- Compute the totient $\varphi(N) = (p - 1)(q - 1)(r - 1)$.
- Choose e coprime to $(p - 1)(q - 1)(r - 1)$.
- Compute $d \equiv e^{-1} \pmod{(p - 1)(q - 1)(r - 1)}$.
- Encrypt: $E(x) = x^e \bmod N$. Decrypt: $D(y) = y^d \bmod N$.

Proof that $D(E(x)) \equiv x \pmod{N}$.

Since $d \equiv e^{-1} \pmod{(p-1)(q-1)(r-1)}$, we have $ed = 1 + k(p-1)(q-1)(r-1)$ for some integer k . Then

$$D(E(x)) = x^{ed} = x^{1+k(p-1)(q-1)(r-1)} = x(x^{k(p-1)(q-1)(r-1)} - 1) + x,$$

so it suffices to show $x(x^{k(p-1)(q-1)(r-1)} - 1) \equiv 0 \pmod{N}$.

Divisibility by p . Two cases:

- $p \nmid x$: By Fermat's Little Theorem, $x^{p-1} \equiv 1 \pmod{p}$. Then

$$x^{k(p-1)(q-1)(r-1)} = (x^{p-1})^{k(q-1)(r-1)} \equiv 1 \pmod{p},$$

$$\text{so } x^{k(p-1)(q-1)(r-1)} - 1 \equiv 0 \pmod{p}.$$

- $p \mid x$: Then x is a factor of the expression, so it is divisible by p .

In both cases $x(x^{k(p-1)(q-1)(r-1)} - 1) \equiv 0 \pmod{p}$.

Divisibility by q and r . By identical symmetric arguments (grouping the exponent as $(x^{q-1})^{k(p-1)(r-1)}$ and $(x^{r-1})^{k(p-1)(q-1)}$ respectively), the expression is also divisible by q and by r .

Combining. The expression is divisible by each of the distinct primes p, q, r , hence by their product $pqr = N$. Therefore

$$x^{ed} - x \equiv 0 \pmod{N} \implies D(E(x)) = x^{ed} \equiv x \pmod{N}. \quad \square$$

Problem 4: RSA with Limited Messages

Alice sends only one of two messages, “Yes” or “No.”

Part (a): How Eve breaks it

Standard RSA encryption is *deterministic*: a fixed message always encrypts to the same ciphertext under a given public key. Since the public key (e, N) is public and Eve knows the message is either “Yes” or “No,” she can precompute both possible ciphertexts:

$$E(\text{“Yes”}) = (\text{“Yes”})^e \bmod N, \quad E(\text{“No”}) = (\text{“No”})^e \bmod N.$$

When she intercepts Alice's actual ciphertext, she compares it against these two precomputed values; the match reveals exactly which message Alice sent. The vulnerability is determinism combined with a tiny message space, which lets Eve enumerate and match without breaking any math.

Part (b): Fixing it with randomness (a one-time pad)

To defeat the attack, Alice injects fresh randomness so that identical messages produce different ciphertexts. One construction:

1. Alice picks a fresh random value r for each transmission.
2. Alice sends Bob two things: the RSA-encryption of r , namely $r^e \bmod N$, together with the message combined with the pad, e.g. $(\text{message} \oplus r)$ (or $(\text{message} + r) \bmod N$).
3. Bob decrypts r^e using his private key to recover r , then undoes the combination to recover the message.

Because r is random and used only once, the same message “Yes” encrypts to a different transmission each time, so Eve can no longer precompute and match. The one-time pad r hides which message was sent by randomizing the ciphertext.